

Pattern-1:

row=5

for i in range(1,row+1):

 a=65

 n=97

 for j in range(1,i+1):

 if i==2 or i==4:

 print(chr(n),end="")

 n+=1

 elif i==1 or i==3 or i==5:

 print(chr(a),end="")

 a+=1

 print()

o/p:

A

ab

ABC

abcd

ABCDE

Pattern-2:

row=5

for i in range(1,row+1):

 a=65

 for j in range(1,i+1):

 if i==2 or i==4:

 print(j,end="")

 elif i==1 or i==3 or i==5:

 print(chr(a),end="")

```
        a+=1
    print()
```

o/p:

```
A
12
ABC
1234
ABCDE
```

Pattern-3:

```
row=5
for i in range(1,row+1):
    for j in range(1,i+1):
        print(j,end="")
    print()
```

o/p:

```
1
12
123
1234
12345
```

Pattern-4:

```
row=5
for i in range(1,row+1):
    n=65
    for j in range(1,i+1):
        print(chr(n),end="")
        n+=1
    print()
```

o/p:

A

AB

ABC

ABCD

ABCDE

Pattern-5:

row=5

```
for i in range(1,row+1):
```

```
    for j in range(1,i+1):
```

```
        print(i,end="")
```

```
    print()
```

o/p:

1

22

333

4444

55555

Nested for loops:

```
1.for i in range(1,4):  
    for j in range(1,11):  
        print(f'{i}*{j} = {i*j}')  
    print("-----")  
o/p:
```

1*1 = 1

1*2 = 2

1*3 = 3

1*4 = 4

1*5 = 5

1*6 = 6

1*7 = 7

1*8 = 8

1*9 = 9

1*10 = 10

2*1 = 2

2*2 = 4

2*3 = 6

2*4 = 8

2*5 = 10

2*6 = 12

2*7 = 14

2*8 = 16

2*9 = 18

2*10 = 20

3*1 = 3

$$3*2 = 6$$

$$3*3 = 9$$

$$3*4 = 12$$

$$3*5 = 15$$

$$3*6 = 18$$

$$3*7 = 21$$

$$3*8 = 24$$

$$3*9 = 27$$

$$3*10 = 30$$

2.for i in range(1,3):

 for j in range(1,3):

 print("**", end="")

 print()

o/p:

**

**

3.for i in range(1,6):

 for j in range(1,1+i):

 print("**",end="")

 print()

o/p:

*

**

4.for i in range(1,6):

 for j in range(1,1+i):

```
print(j,end="")
```

```
print()
```

o/p:

1

12

123

1234

12345

5. for i in range(1,6):

```
    for j in range(1,1+i):
```

```
        print(i,end="")
```

```
    print()
```

o/p:

1

22

333

4444

55555

6. row=5

```
for i in range(1,row+1):
```

```
    for j in range(1,row+1-i):
```

```
        print(j,end="")
```

```
    print()
```

1234

123

12

1

7.row=5

n=1

```
for i in range(1,row+1):  
    for j in range(1,i+1):  
        print(n,end="")  
        n+=1  
    print()
```

1.

row=5

```
for i in range(1,row+1):  
    for j in range(1,i+1):  
        print(j, end="")  
    print()
```

o/p:

1

12

123

1234

12345

2.

row=5

```
for i in range(1,row+1):  
    for j in range(1,i+1):  
        print(i, end="")  
    print()
```

o/p:

1

22

333

4444

55555

3.

row=5

n=1

for i in range(1,row+1):

 for j in range(1,i+1):

 print(n, end="")

 n+=1

 print()

o/p:

1

23

456

78910

1112131415

4.

row=5

n=1

for i in range(row,0,-1):

 for j in range(1,i+1):

 print(j, end="")

 print()

o/p: 12345

1234

123

12

1

5.

row=5


```
n=1
for i in range(row,0,-1):
    for j in range(1,i+1):
        print(i, end="")
```

```
print()
```

o/p:

55555

4444

333

22

1